



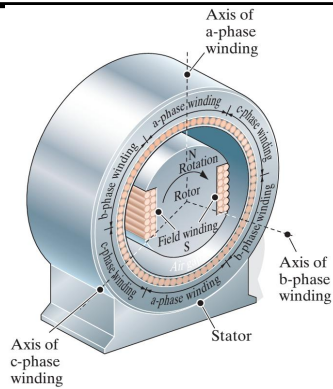
Lecture 12

Transformers/Three-Phase Circuits



Outline

- Balanced Three-Phase System (Wye - Wye)
 - Balanced sources
 - Balanced loads
- Circuit analysis (Wye - Wye)
 - Phase voltage/current
 - Line voltage/current



Generating Station
11, 11.5, 12 or 13kV



Step-Up Transformer
(Power Transformer)

220 or
500 kV

Primary Transmission

3 Phase 3 Wire System

<http://electricaltechnology.org/>

Secondary Transmission

3 Phase 3 Wire System

Step-Down Transformer

220 or 500 kV
/ 132 or 66 kV

<http://electricaltechnology.org/>

132 or 66 kV / 11 kV

Step-Down Transformer

11kV/ 400 & 230 V
3 Phase 4 Wire System

Primary Distribution

3 Phase 3 Wire System

Distribution Transformer

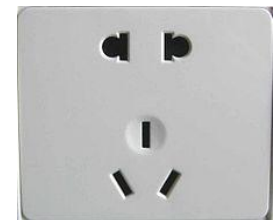
Secondary Distribution

<http://electricaltechnology.org/>

Design By:
Engr Wasim Khan

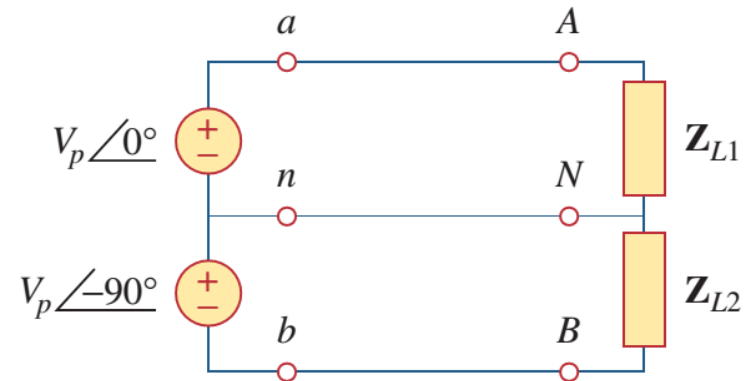
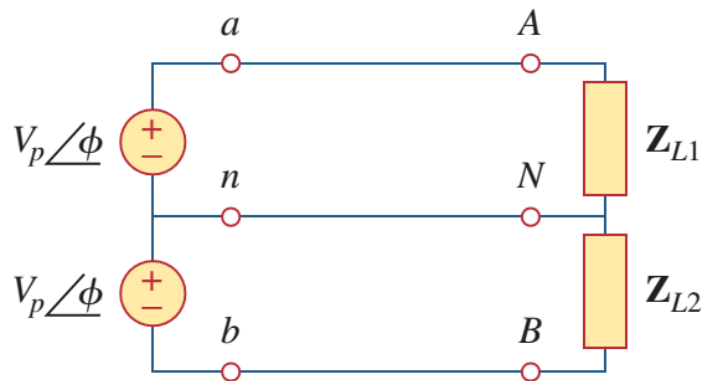
Residential Consumers

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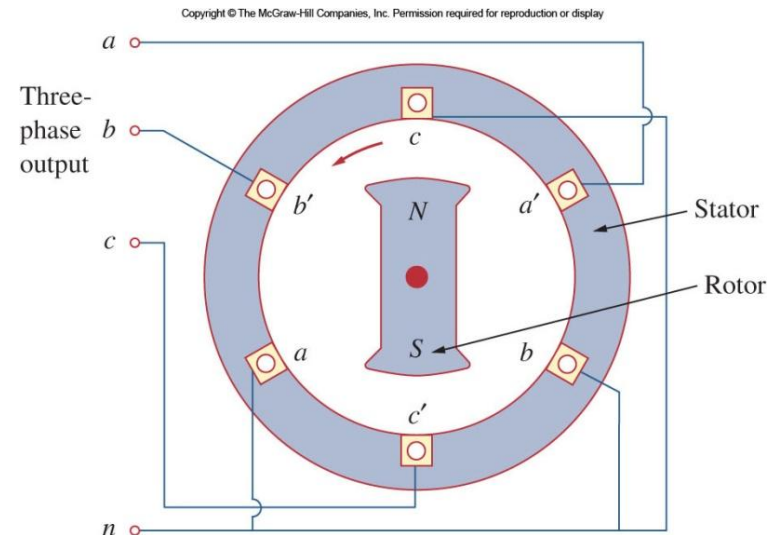
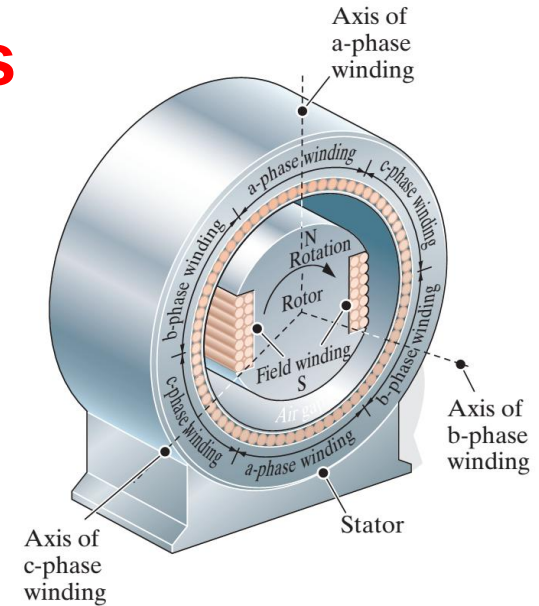
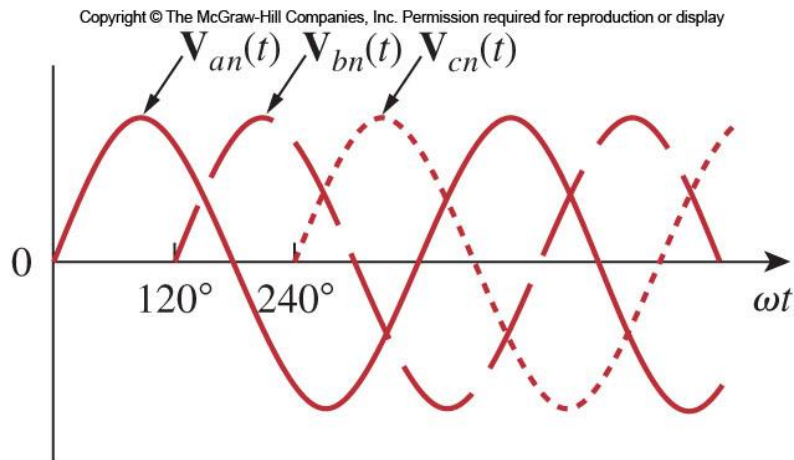
Single Phase vs. Polyphase

- Households have single-phase power supply
 - This typically in a three wire form
- Circuits that operate at the same frequency but with multiple sources at different phases are called polyphase.



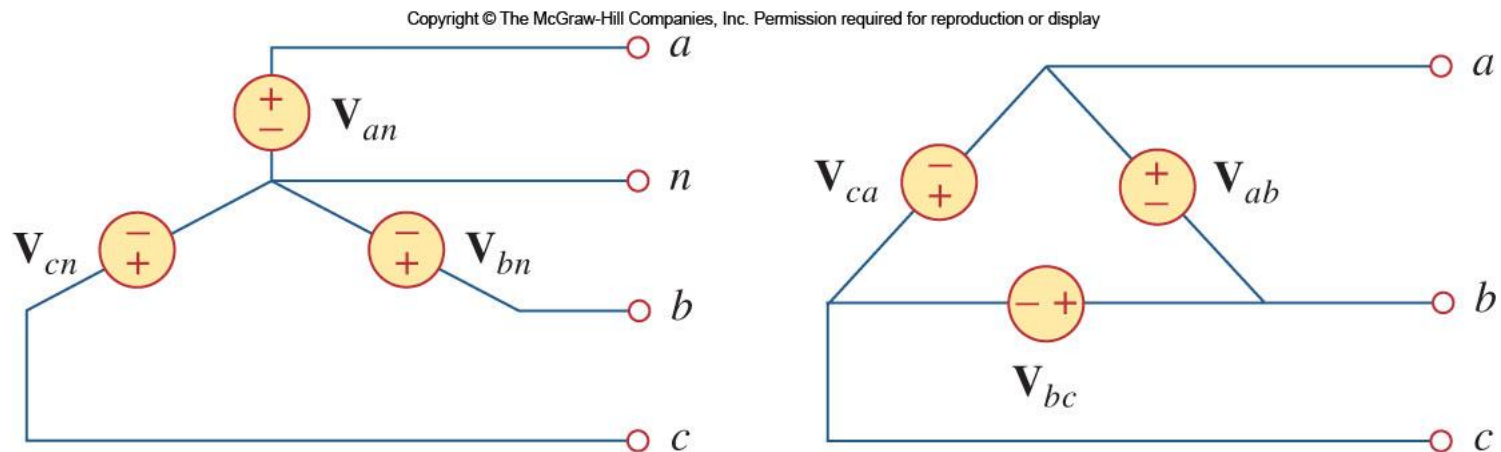
Balanced Three-Phase Sources

- Three phase voltages are typically produced by a three-phase AC generator.
- The output voltages look like below.



Connecting the Sources

- Three phase voltage sources can be connected to the loads by either three or four wire configurations.
 - Three-wire configuration accomplished by Delta connected source.
 - Four-wire system accomplished using a Y connected source.

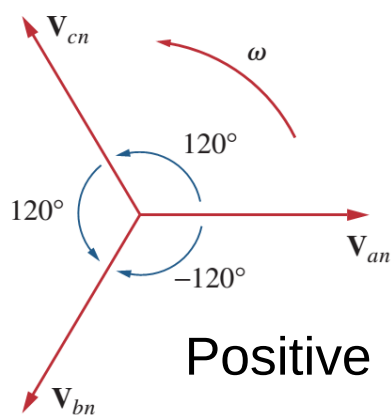


Balanced Sources

- A Y connected source is said to be balanced when

$$V_{an} + V_{bn} + V_{cn} = 0$$

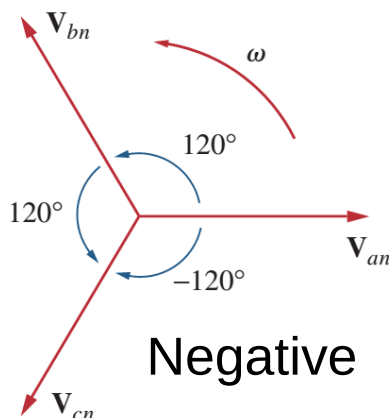
- Two sequences for the phases:



$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

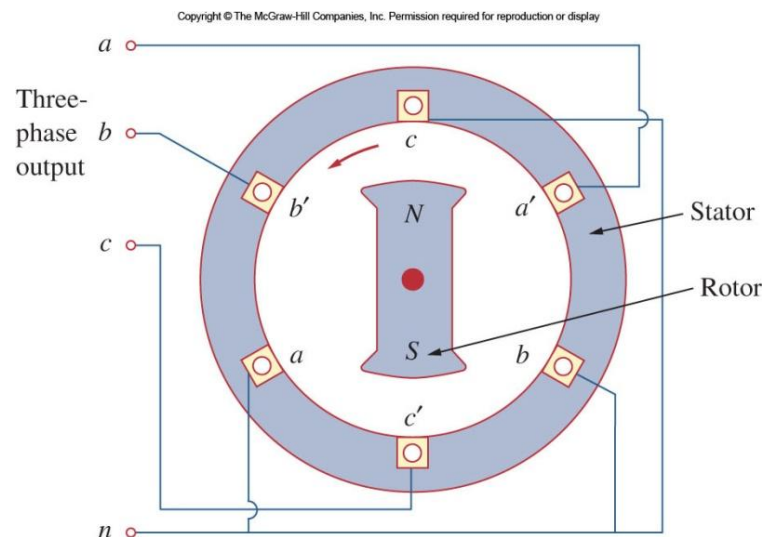
$$V_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$



$$V_{an} = V_p \angle 0^\circ$$

$$V_{cn} = V_p \angle -120^\circ$$

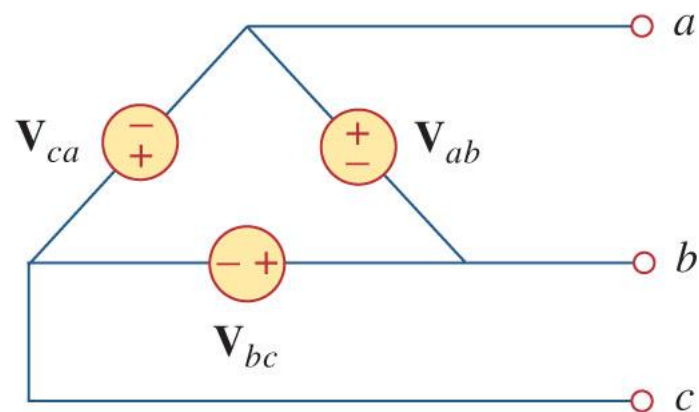
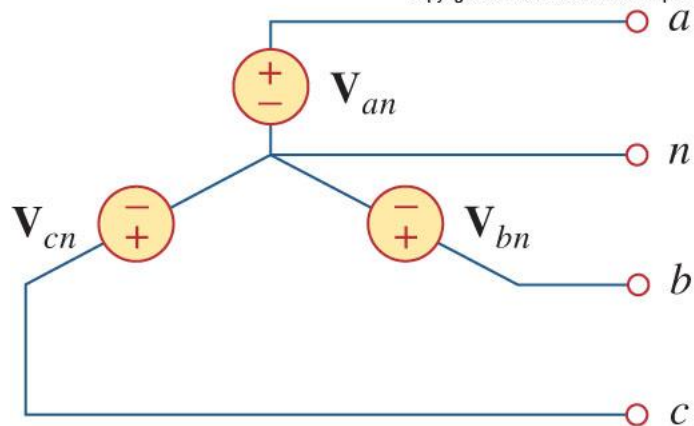
$$V_{bn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$



$$|V_{an}| = |V_{bn}| = |V_{cn}|$$

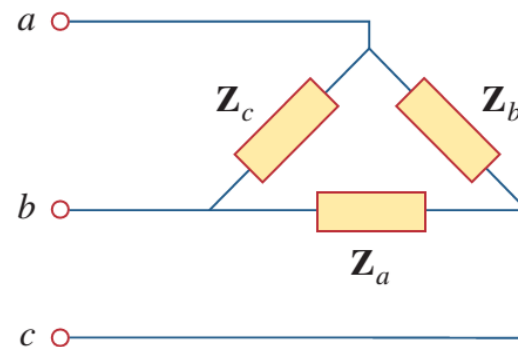
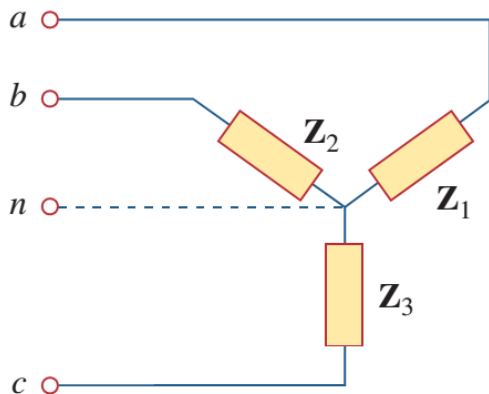


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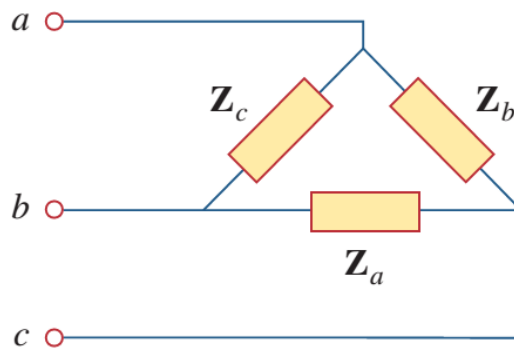
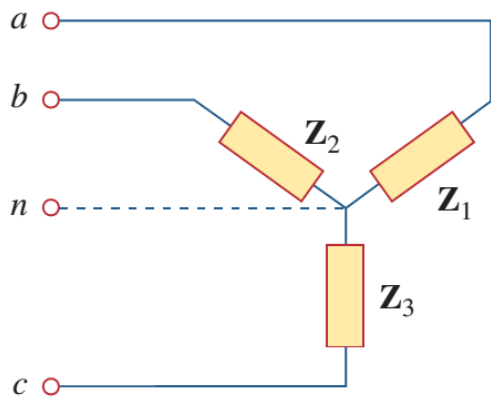


Balanced Loads

- Similar to the source, a balanced load is one that has the same impedance presented to all three voltage sources.
- They may also be connected in either Delta or wye
 - For a balanced wye connected load: $Z_1 = Z_2 = Z_3 = Z_Y$
 - For a balanced delta connected load: $Z_a = Z_b = Z_c = Z_\Delta$



- The load impedance per phase for the two load configurations can be interchanged.

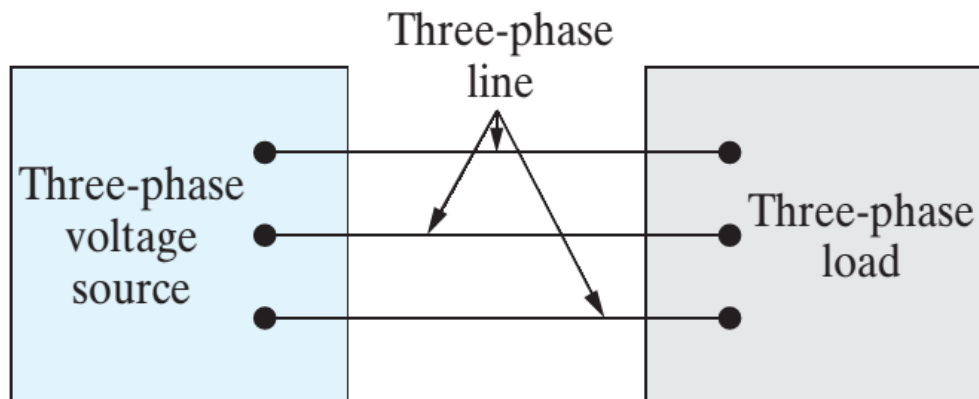




Outline

- Balanced Three-Phase System (Wye - Wye)
 - Balanced sources
 - Balanced loads
- Circuit analysis (Wye - Wye)
 - Phase voltage/current
 - Line voltage/current

Source-Load configurations

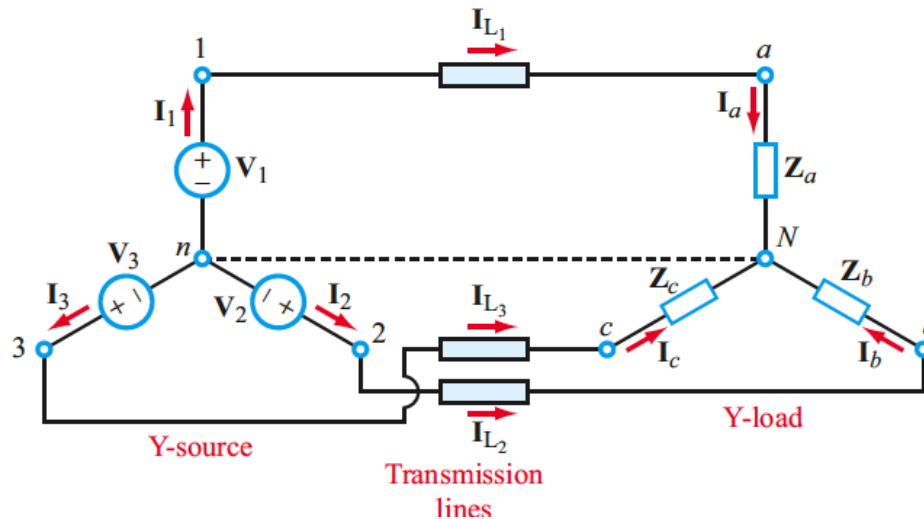


Source	Load
Y	Y
Y	Δ
Δ	Y
Δ	Δ



Source-Load Configurations

Y-Y



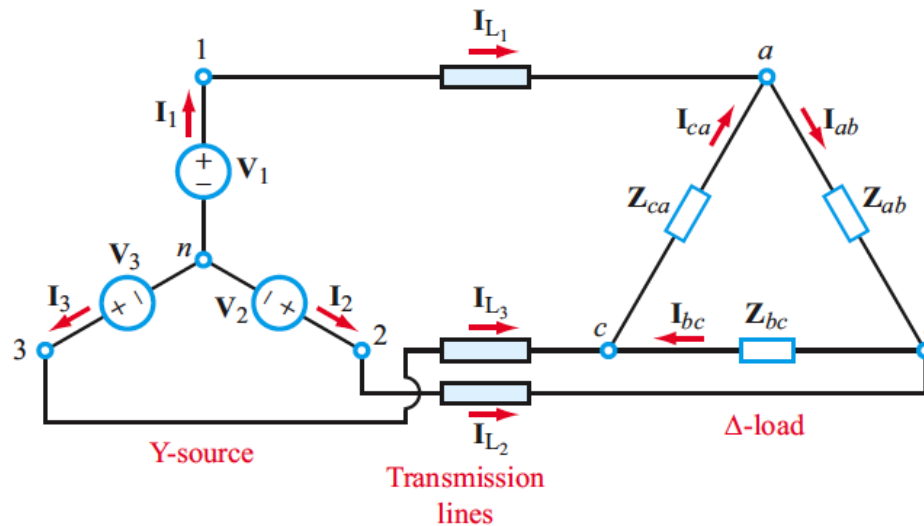
Load Phase Currents

I_a, I_b, I_c
(same as line currents
 $I_{L1}, I_{L2}, \text{ and } I_{L3}$)

Load Phase Voltages

V_{aN}, V_{bN}, V_{cN}

Y-Delta



Load Phase Currents

I_{ab}, I_{bc}, I_{ca}

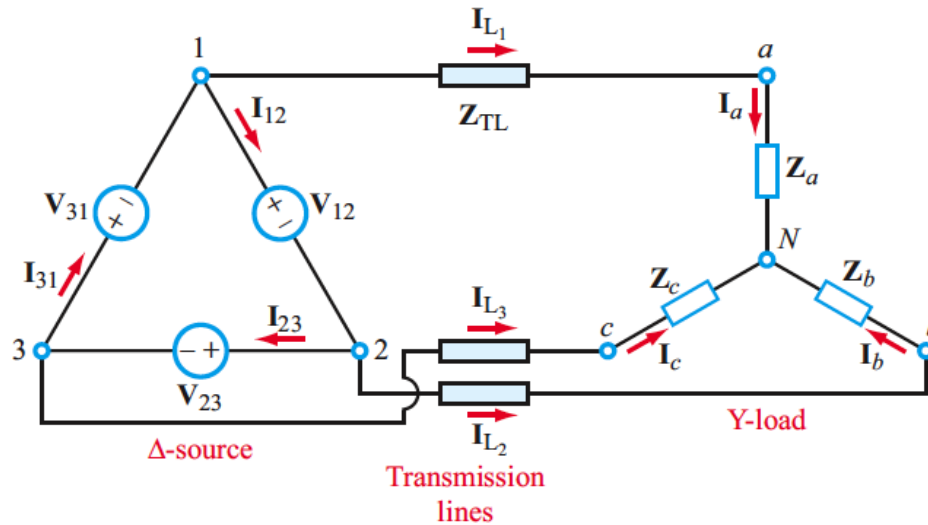
Load Phase Voltages

V_{ab}, V_{bc}, V_{ca}



Source-Load Configurations

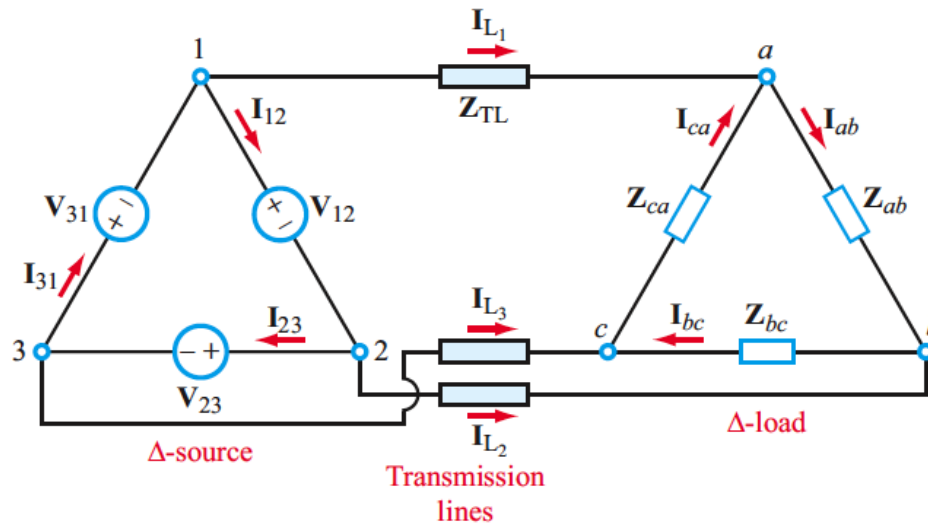
Delta-Y



Load Phase Currents
 I_a, I_b, I_c
(same as line currents
 $I_{L1}, I_{L2}, \text{ and } I_{L3}$)

Load Phase Voltages
 V_{aN}, V_{bN}, V_{cN}

Delta-Delta



Load Phase Currents
 I_{ab}, I_{bc}, I_{ca}

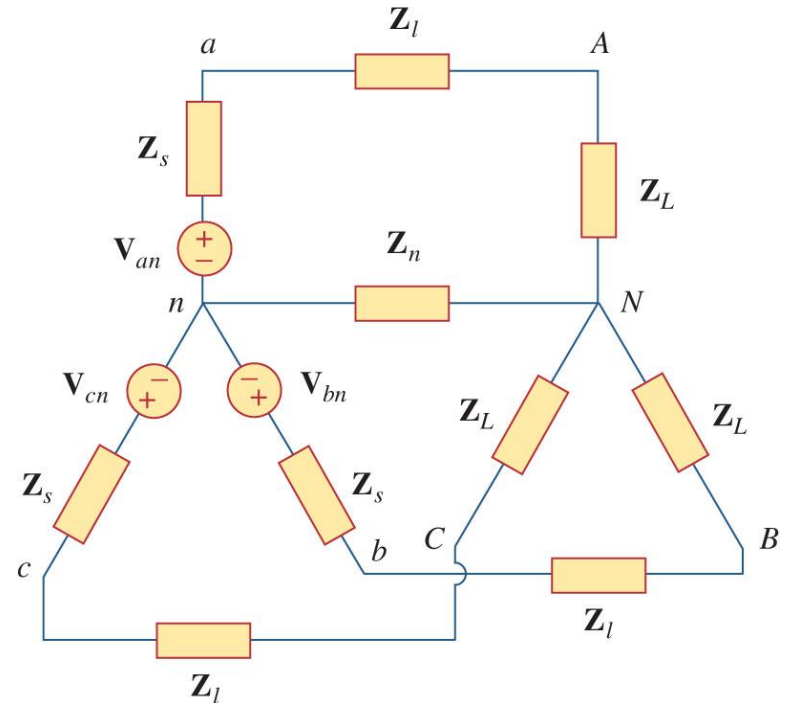
Load Phase Voltages
 V_{ab}, V_{bc}, V_{ca}
(same as source voltages
if Z_{TL} is negligible)

Balanced Y-Y connection

- Any three-phase system can be reduced to an equivalent Y-Y system.
- The load impedances Z_y will be assumed to be balanced.
 - This can be the source Z_s , line Z_l and load Z_L together.

$$Z_Y = Z_s + Z_\ell + Z_L$$

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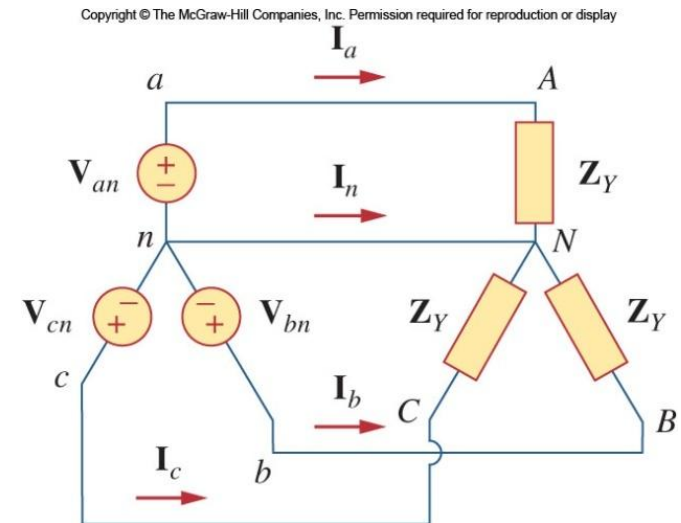
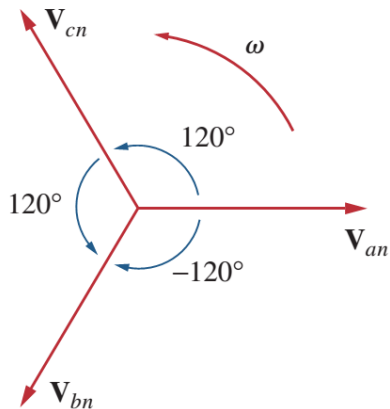
Line-to-Line Voltage

- Use the positive sequence:

$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ \quad V_{cn} = V_p \angle +120^\circ$$

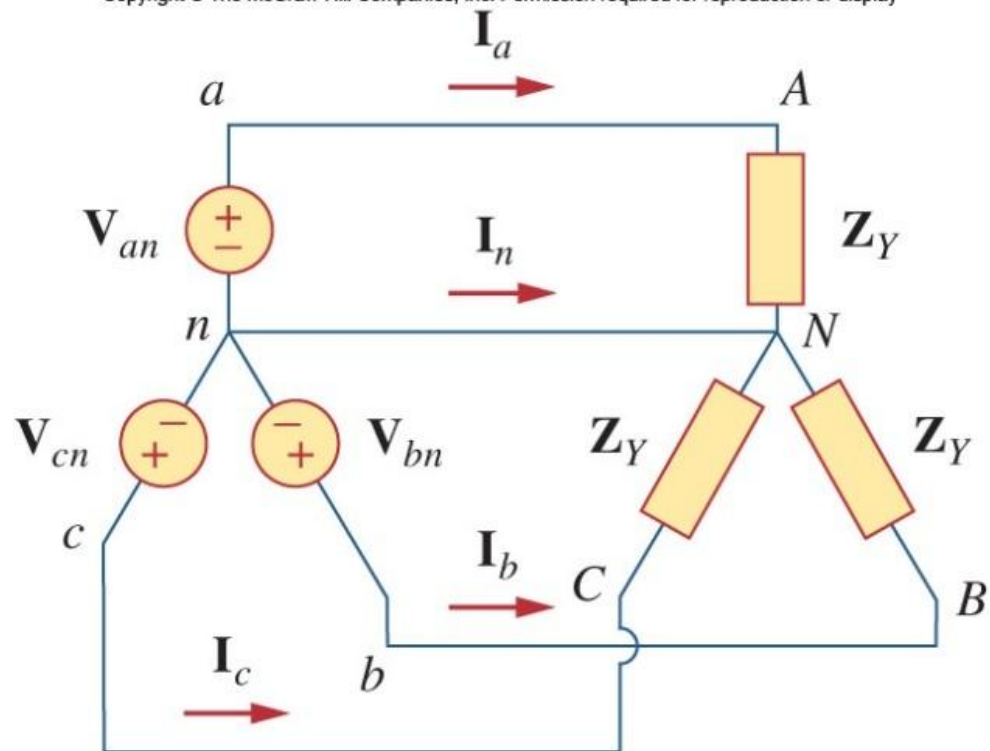
The line to line (or line in short) voltages:





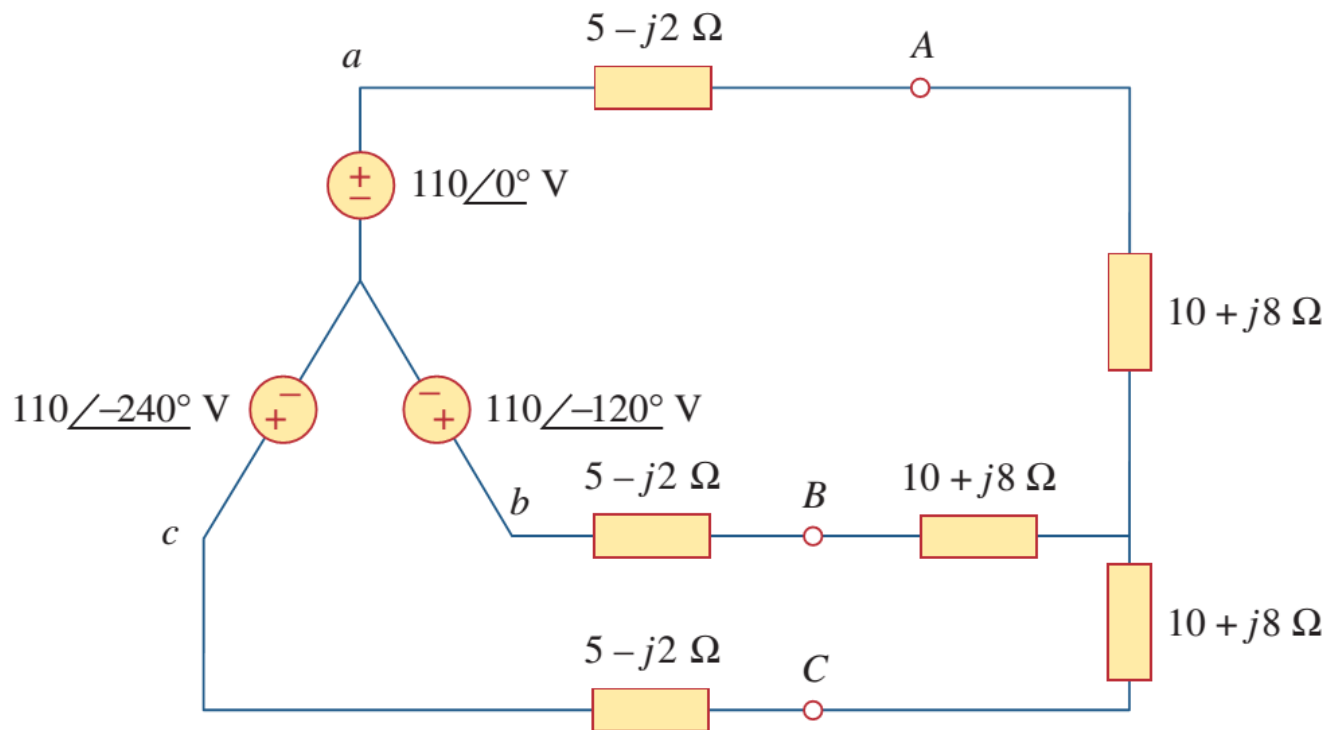
Line Currents

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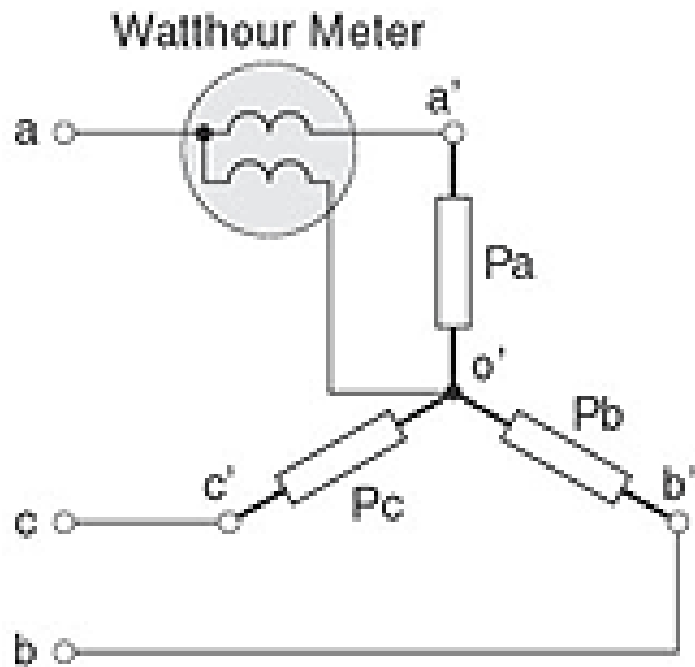


Example

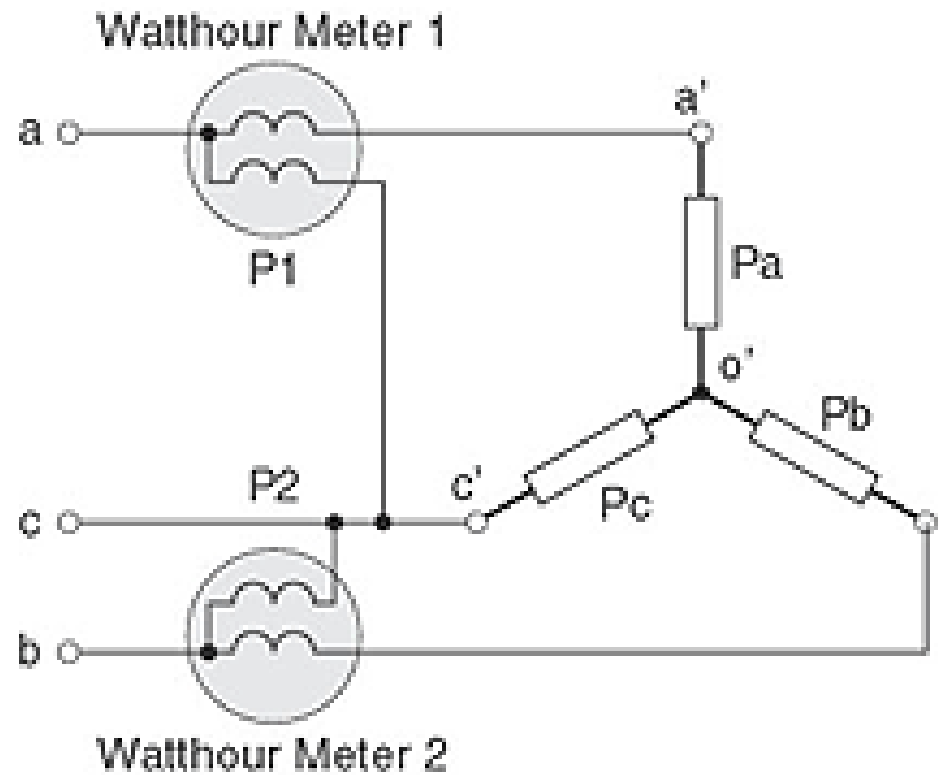
- Calculate the line currents.



Power



$$P = P_a + P_b + P_c \equiv 3 P_a$$



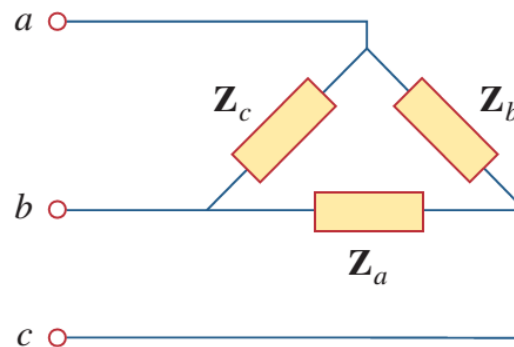
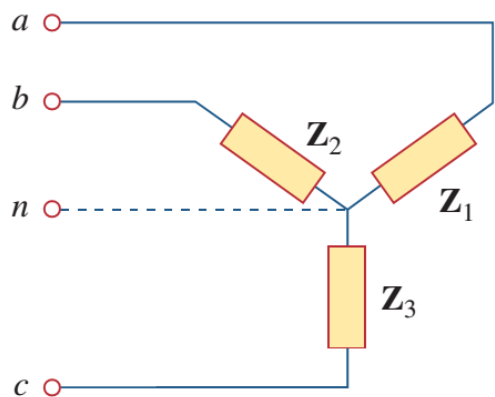
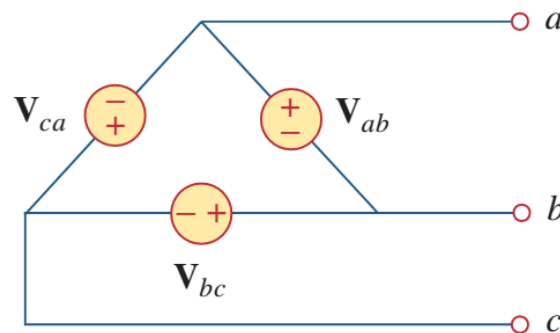
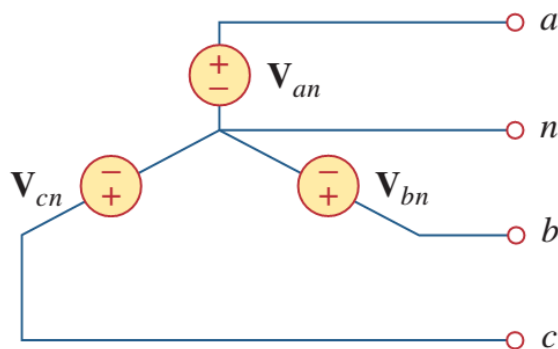
$$P = P_a + P_b + P_c = P_1 + P_2$$



Y-Y	$\mathbf{V}_{an} = V_p \angle 0^\circ$ $\mathbf{V}_{bn} = V_p \angle -120^\circ$ $\mathbf{V}_{cn} = V_p \angle +120^\circ$ Same as line currents	$\mathbf{V}_{ab} = \sqrt{3}V_p \angle 30^\circ$ $\mathbf{V}_{bc} = \mathbf{V}_{ab} \angle -120^\circ$ $\mathbf{V}_{ca} = \mathbf{V}_{ab} \angle +120^\circ$ $\mathbf{I}_a = \mathbf{V}_{an} / \mathbf{Z}_Y$ $\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a \angle +120^\circ$
Y-Δ	$\mathbf{V}_{an} = V_p \angle 0^\circ$ $\mathbf{V}_{bn} = V_p \angle -120^\circ$ $\mathbf{V}_{cn} = V_p \angle +120^\circ$ $\mathbf{I}_{AB} = \mathbf{V}_{AB} / \mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{BC} / \mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{CA} / \mathbf{Z}_\Delta$	$\mathbf{V}_{ab} = \mathbf{V}_{AB} = \sqrt{3}V_p \angle 30^\circ$ $\mathbf{V}_{bc} = \mathbf{V}_{BC} = \mathbf{V}_{ab} \angle -120^\circ$ $\mathbf{V}_{ca} = \mathbf{V}_{CA} = \mathbf{V}_{ab} \angle +120^\circ$ $\mathbf{I}_a = \mathbf{I}_{AB} \sqrt{3} \angle -30^\circ$ $\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a \angle +120^\circ$
Δ-Δ	$\mathbf{V}_{ab} = V_p \angle 0^\circ$ $\mathbf{V}_{bc} = V_p \angle -120^\circ$ $\mathbf{V}_{ca} = V_p \angle +120^\circ$ $\mathbf{I}_{AB} = \mathbf{V}_{ab} / \mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{bc} / \mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{ca} / \mathbf{Z}_\Delta$	Same as phase voltages $\mathbf{I}_a = \mathbf{I}_{AB} \sqrt{3} \angle -30^\circ$ $\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a \angle +120^\circ$
Δ-Y	$\mathbf{V}_{ab} = V_p \angle 0^\circ$ $\mathbf{V}_{bc} = V_p \angle -120^\circ$ $\mathbf{V}_{ca} = V_p \angle +120^\circ$ Same as line currents	Same as phase voltages $\mathbf{I}_a = \frac{V_p \angle -30^\circ}{\sqrt{3}\mathbf{Z}_Y}$ $\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a \angle +120^\circ$



Y and Δ , Which One Better?



<http://www.allaboutcircuits.com/textbook/alternating-current/chpt-10/three-phase-y-delta-configurations/>

4. (18 points in total) A three-phase phasor domain circuit is shown in Figure P4.

(a) (2 points) For the source part, is the connection delta or Y? For the load part, is the connection delta or Y?

(b) (2 points) Is the system a balanced three phase system?

(c) (8 points) Find the load voltages $\tilde{V}_{an}$, $\tilde{V}_{bn}$ and $\tilde{V}_{cn}$ and load currents $\tilde{I}_a$, $\tilde{I}_b$ and $\tilde{I}_c$.

(d) (6 points) For each phase, find the average power (or real power), reactive power, complex power absorbed by the load of each phase. Find the power factor of each phase and determine whether it is a leading or lagging power factor.

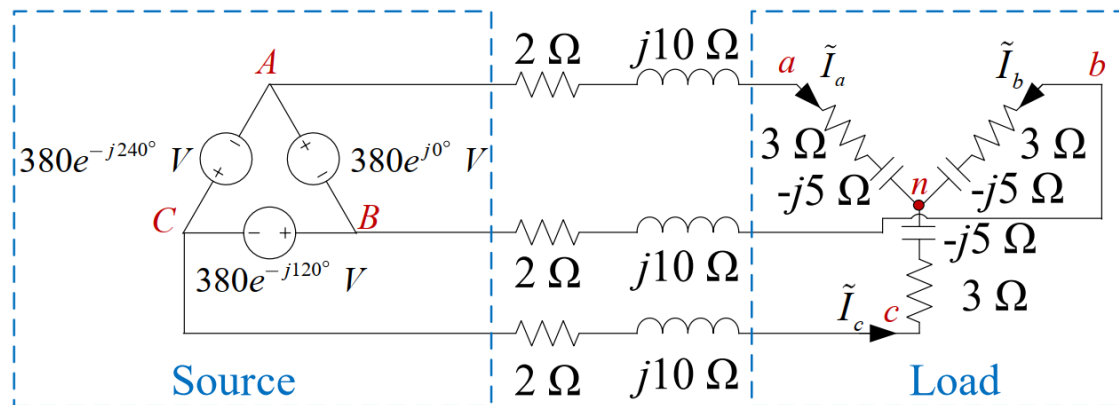


Figure P4